

WEEK 2 Day 9

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Cleaning Proposal & Mentor Review

What We Found — Data Quality

- 208 CSV files loaded across 6 temperature conditions (−20 °C to 40 °C)
- Approximately 4.95 million measurement rows combined into one dataset
- 5,876 duplicate rows (0.12 %) identified — safe to drop
- 3 files contain non-positive voltage (physically impossible):
590_Charge16.csv, 552_Charge16.csv, 557_Charge10.csv
- No SOC column exists in the raw data → SOC must be derived
- Temperature is a real signal driver (see plots in Day 8 notebook)

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Missing Values:-
Total missing values: 4954079

Duplicate Rows:-
Total duplicate rows: 5876

T  5433
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	Temperature	Number of Files	Total Rows	Total Missing Values	Total Duplicate Rows
0	0degC	32	850622	850622	232
1	10degC	32	925533	925552	456
2	25degC	36	1045475	1044620	287
3	40degC	35	692629	692629	269
4	n10degC	38	829826	829826	4398
5	n20degC	35	610830	610830	234

Temperature summary saved:
/content/drive/MyDrive/BATTERY_SOC_PROJECT/SOC DATA SET/Day_7_Reports/Day7_Temperature_Summary.csv

Proposed Cleaning & Split Logic

1. STANDARDIZE — strip 28-row metadata, unify column names, force numeric dtypes
2. CLEAN — drop exact duplicates; FLAG (do not auto-delete) non-positive voltage rows
3. OUTLIERS — apply IQR on Current & Temperature → flag only, review before removal
4. DERIVE SOC — Coulomb counting using Capacity + Current integration
5. SPLIT LOGIC — split BY CYCLE, not by row → prevents train/test leakage

Representative Outliers & Invalid Records

INVALID VOLTAGE (3 files)

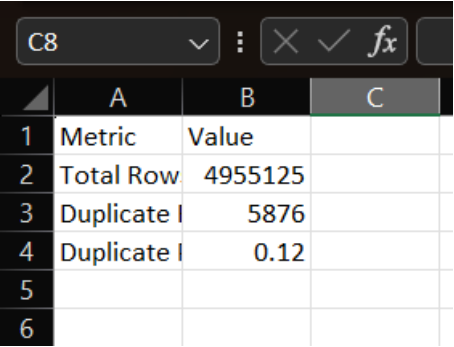
- 590_Charge16.csv | 552_Charge16.csv | 557_Charge10.csv
- Cause: sensor dropout at test start
- Decision: FLAG, do not auto-delete

DUPLICATE ROWS

- 5,876 rows across 208 files (0.12 % of total)
- Highest concentration: n10degC folder (4,398 rows)
- Decision: SAFE TO DROP (exact duplicates)

STATISTICAL OUTLIERS

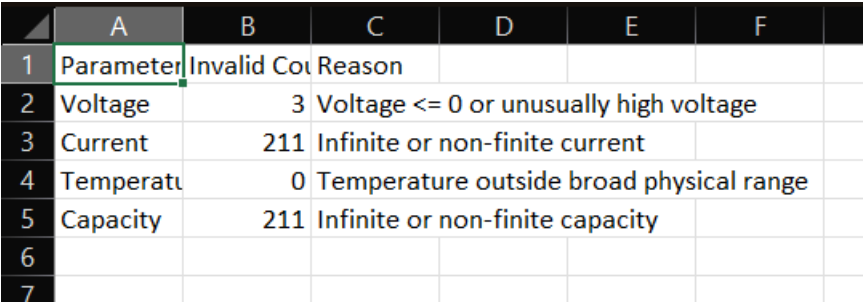
- Current: 6.75 % flagged (expected — high-load pulses)
- Voltage: 0.00 % | Temperature: 0.00 % | Capacity: 0.21 %
- Decision: Investigate before removal



A screenshot of a spreadsheet window showing a table with 3 columns (A, B, C) and 6 rows. The table contains summary statistics for duplicate reports.

	A	B	C
1	Metric	Value	
2	Total Row	4955125	
3	Duplicate I	5876	
4	Duplicate I	0.12	
5			
6			

Duplicate reports



A screenshot of a spreadsheet window showing a table with 6 columns (A, B, C, D, E, F) and 7 rows. The table lists parameters that are invalid, the count of occurrences, and the reason for invalidity.

	A	B	C	D	E	F
1	Parameter	Invalid Count	Reason			
2	Voltage	3	Voltage <= 0 or unusually high voltage			
3	Current	211	Infinite or non-finite current			
4	Temperature	0	Temperature outside broad physical range			
5	Capacity	211	Infinite or non-finite capacity			
6						
7						

Invalid values report

Train / Test Split Logic

WHY RANDOM SPLIT FAILS HERE

- Rows within one discharge cycle are highly correlated
- Random split → same cycle appears in both train and test
- Result: inflated accuracy, poor real-world performance

PROPOSED SPLIT STRATEGY

- Split BY CYCLE ID (column: 'Cycle')
- Each cycle belongs entirely to either train OR test
- Target ratio: 80 / 20 (by cycle count, not row count)

FURTHER ISOLATION

- Hold out one temperature condition entirely (e.g., 25 °C)
- Tests model generalisation to unseen conditions

Expected Dataset After Cleaning

BEFORE CLEANING

- 208 files | ~4.95 M rows | 15 columns
- Inconsistent parsing across files
- No SOC target

AFTER CLEANING

- Consistent schema across all files
- Duplicates removed (~5,870 rows)
- Invalid rows flagged with a 'reason' column
- SOC column derived and validated
- Train / Val / Test split saved as 3 separate files

DELIVERABLES FOR DAY 10

- clean_dataset.parquet
- train.parquet / val.parquet / test.parquet
- cleaning_report.csv (row-by-row trace of drops and flags)

Summary & Approval Request

WHAT WE'VE DONE

- Inspected all 208 files (~4.95 M rows)
- Identified duplicates, invalid voltages, statistical outliers
- Confirmed no SOC column → proposed derivation method
- Designed leakage-safe split strategy

WHAT WE NEED FROM YOU

1. Approve cleaning plan (5 steps on Slide 3)
2. Decide: keep or remove the 3 bad-voltage files?
3. Confirm SOC method: Coulomb counting vs OCV
4. Approve cycle-aware train/test split

NEXT STEP (DAY 10)

- Implement cleaning script | Generate clean + split dataset | Begin feature engineering

Thank You